

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: Sec 1. Introduction To Python

Lecture: 3. Programming Languages

Section 1: Lecture Summary

This lecture introduces the concept of programming languages by first explaining the importance of language for communication, especially for giving instructions or procedures. It draws an analogy between human languages and machine languages, noting that computers understand only binary machine language (zeros and ones), which is very different from human languages like English. Hence, humans cannot program computers directly in natural language.

To bridge this gap, translators like compilers and interpreters are used to convert human-readable programming languages into machine language. Python is described as a hybrid language that uses both compilation and interpretation to achieve this translation. The lecture also distinguishes programming languages into two broad categories: low-level (machine and assembly languages) and high-level (languages like C, C++, Python, Java). High-level languages are easier for humans to learn and use to write programs since they are more abstract and closer to human language.

Overall, the lecture explains why programming languages exist, their role as translators between humans and machines, and categorizes types of programming languages to provide foundational knowledge necessary to learn Python.

Section 2: Key Concepts and Explanations

- Purpose of Language in Communication

- Language enables communication between two parties.
- Language is vital for sharing ideas, stories, and most importantly, giving instructions (procedures).

- Programming is essentially the process of giving a set of instructions (a program) to the computer.

- Human Language vs Machine Language

- Humans use natural languages (e.g., English) to communicate.
- Computers understand only machine language, a binary code of zeros and ones.
- Direct communication between human language and machine language is impossible without translation.

- Role of Translators in Programming

- A translator converts instructions from one language to another.
- Just like a human translator bridges communication between English and Chinese speakers, compiler and interpreter translate programming instructions into machine language.
- **Compiler** converts the entire program into machine code before execution.
- **Interpreter** converts and executes code line-by-line.
- Python uses both, making it a hybrid language.

- Programming Languages

- Languages used by programmers to write instructions for machines.
- Must be translated to machine language by compilers or interpreters.
- Examples: C, C++, Java, Python.

- Types of Programming Languages

- Low-Level Languages

- Close to machine language.
- Examples: Machine language (binary), Assembly language.
- Difficult for humans to learn and use.

- High-Level Languages

- Easier for humans to learn.
- Closer to natural language but still must be translated.
- Categories:
 - Compiler-based: C, C++ (compiled programs).
 - Interpreter-based: JavaScript, PHP (interpreted line-by-line).
 - Hybrid: Python, Java, C# (combination of compiler and interpreter).

- Summary

- Programming languages allow humans to communicate instructions to machines.
- Translators (compilers/interpreters) enable this communication by converting high-level languages into machine language.
- Understanding these concepts is foundational before learning Python.

Section 3: Example Code and Use Cases

The lecture transcript did not include explicit code examples to reconstruct. However, based on the explained concepts, here is a simple illustrative example related to Python as a hybrid language, showing a Python program and its conceptual translation by the interpreter/compiler:

Python program to print "Hello, World!"

```
print("Hello, World!")
```

Explanation:

- The programmer writes code in Python, which is English-like but structured syntax.
- The Python interpreter line-by-line translates and executes this instruction to print the text on the screen.

- Under the hood, a compiler component converts Python code into intermediate bytecode, then the interpreter executes it on the machine.

Section 4: Key Takeaways

- Language is essential for communication and for giving instructions (programming is instruction-giving).

- Computers understand only machine language (binary), which humans cannot write directly.

- Translators like compilers and interpreters convert human-readable programming languages into machine language.

- Python is a hybrid programming language using both compiler and interpreter techniques.

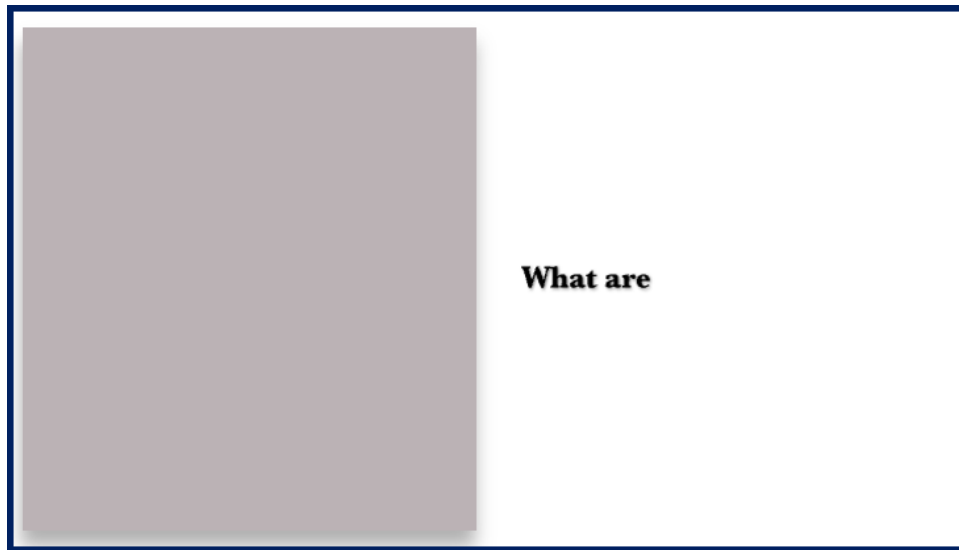
- Programming languages are classified into low-level (machine, assembly) and high-level (C, Python, Java).

- High-level languages are easier to learn and use, but require translation to machine language before execution.

- Understanding the purpose and types of programming languages is key before learning Python programming.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.





What are **Programming Languages** ?



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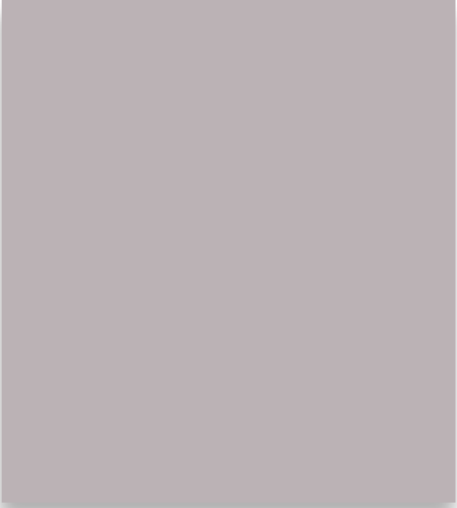
What are Programming Languages ?

- Why we use Languages ?

What are Programming Languages ?

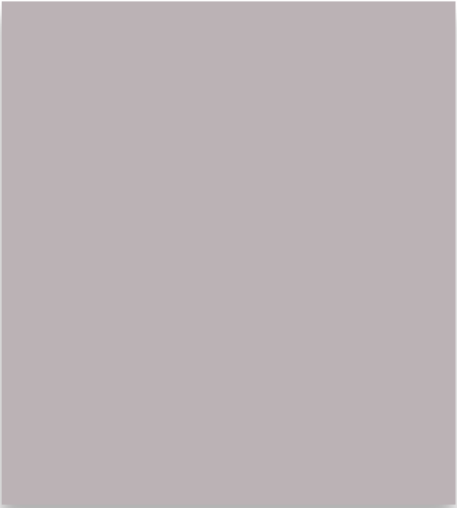
- Why we use Languages ?

- Translation



What are Programming Languages ?

- Why we use Languages ?
- Translation
- Types of Programming Languages



Why we use Languages ?

Why we use Languages ?

Why we use Languages ?



Why we use Languages ?

Language



Why we use Languages ?

Language



Why we use Languages ?

Language



Sharing Idea

Why we use Languages ?

Language



Sharing Story

Why we use Languages ?

Language



Gossip

Why we use Languages ?

Language



Instructions

Why we use Languages ?

Language



Instructions

Why we use Languages ?

Language

English

English

Instructions

Why we use Languages ?

Language

English

Instructions

Why we use Languages ?

Language



Instructions

Why we use Languages ?

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Why we use Languages ?

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Instructions

Why we use Languages ?

Language



Instructions



Translation



Translation



Translation



Translation



Translation

English

Translator

福



Language



Instructions

Translation

English

Translator (Chinese)

福



Language



Instructions

